universität freiburg

Machine Learning: Organizational Matters

Dr. Simon Bultmann, Prof. Dr. Abhinav Valada

Winter Semester 2026-2027

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robot learning



Machine Learning Course

- Scope:
  - Introductory Machine Learning
  - Mostly supervised learning
  - Building blocks for forthcoming courses on Deep Learning, Reinforcement Learning, AutoML
- Requirements:
  - Basic Math, Calculus, Linear Algebra
  - Algorithms
  - Statistics
  - Programming in Python
- Credits and evaluation:
  - ECTS: **6** credits
  - Evaluation: **Closed-book, written** examination
  - Studienleistung: **Passing the exam**

Team



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Registration and Access

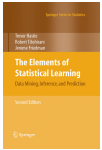


- Register in HISinOne and follow the course on ILIAS
- Registration:
  - HISinOne:
    - Course ID: **11LE13V-1153** and **11LE13Ü-1153**
  - ILIAS:
    - Course Name: **Machine Learning WS2026/2027**
    - Location: “Lehrveranstaltungen im WS 2026/2027 → Technische Fakultät → Studiengang Informatik/Angewandte Informatik → Master - Seminare PO-Version 2011 → Kognitive technische Systeme → Machine Learning WS2026/2027”
    - Direct link: `https://ilias.uni-freiburg.de/goto.php/crs/4353757`
    - Password: let_me_in_ML2026!

Traditional classroom

- The lecture is held in presence on Mondays at 10:00 in HS 00 026 (Building 101)
- The exercise tutorial will be held in presence on Friday at 14:00 in HS 00 006 (Building 82 – “Kino-Hörsaal”)
- Students can ask questions any-time via ILIAS

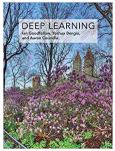
Suggested Books



- The Elements of Statistical Learning
  by Hastie, Tibshirani and Friedman
  - Available online for free:
    `https://web.stanford.edu/~hastie/ElemStatLearn/`



- Pattern Classification
  by Duda, Hart and Stork



- Deep Learning
  by Goodfellow, Bengio and Courville
  - HTML version available online for free:
    `http://www.deeplearningbook.org/`

Related Courses at Uni-Freiburg

- Foundations of Deep Learning
  - By Frank Hutter, Abhinav Valada
  - More advanced course that builds on this current course
- Advanced Deep Learning (ADL)
  - By Abhinav Valada
  - More advanced course that builds on this current course & Deep Learning

Related Courses at Uni-Freiburg

- Reinforcement Learning
  - By Joschka Boedecker
  - More advanced course that builds on this current course
- Automated Machine Learning (AutoML)
  - By Frank Hutter
  - More advanced course that builds on this current course & Deep Learning

Course Schedule

	Lecture Nr	Topic	Lecture	Exercise
⇒	Lecture 1	Introduction	19.10.26	30.10.26
	Lecture 2	Linear Methods	26.10.26	06.11.26
	Lecture 3	Principles of Regularization	02.11.26	13.11.26
	Lecture 4	Support Vector Machines	09.11.26	20.11.26
	Lecture 5	Decision Trees	16.11.26	27.11.26
	Lecture 6	Neural Networks (I)	23.11.26	04.12.26
	Lecture 7	Neural Networks (II)	30.11.26	11.12.26
	Lecture 8	Ensembles	07.12.26	18.12.26
	Lecture 9	Gradient-Boosted Trees	14.12.26	15.01.27
	Lecture 10	Hyper-parameter Tuning	11.01.27	22.01.27
	Lecture 11	Fighting Overfitting	18.01.27	29.01.27
	Lecture 12	Error Metrics	25.01.27	05.02.27
	Lecture 13	Round-up / Exam Q & A	01.02.27	-

Assignments Schedule

	Assign.	Topic	Release	Due
⇒	Assign. 1	Introduction	19.10.26	27.10.26
	Assign. 2	Linear Methods	26.10.26	03.11.26
	Assign. 3	Principles of Regularization	02.11.26	10.11.26
	Assign. 4	Support Vector Machines	09.11.26	17.11.26
	Assign. 5	Decision Trees	16.11.26	24.11.26
	Assign. 6	Neural Networks (I)	23.11.26	01.12.26
	Assign. 7	Neural Networks (II)	30.11.26	08.12.26
	Assign. 8	Ensembles	07.12.26	15.12.26
	Assign. 9	Gradient-Boosted Trees	14.12.26	12.01.27
	Assign. 10	Hyper-parameter Tuning	11.01.27	19.01.27
	Assign. 11	Fighting Overfitting	18.01.27	26.01.27
	Assign. 12	Error Metrics	25.01.27	02.02.27

Assignments

- Guidelines:
  - Team work to foster collaboration between students
  - Not graded, yet **crucial** for exam preparation
  - Mix of analytical and programming tasks
  - Released on each lecture **Monday** and due on the following week's **Tuesday**
    - Tutors support assignment-related questions in this period
- Next step:
  - Assemble your teams by chatting at the "Create a team" forum at ILIAS
  - Start working on the first assignment **released today and due on 27.10.26; 23h59**
  - When uploading your solution (voluntary), create your team in the ILIAS exercise submission system

Uploading exercise sheet solutions (voluntary)

- Use the forum **“Create your team”** at ILIAS to scout for team members
  - Suggested team size: 3–4 students
- Solutions must be submitted **as a group**; we cannot provide feedback for individual submissions
- Teams are created via the **self-organized team submission** in ILIAS **when uploading your solution**
- Upload your solutions as “`Exercise_XX_Team_ABC[D].zip`”, with A, B, C, D the initials of your team members
  - Submissions are **voluntary**
  - If possible, upload in machine-readable format (e.g. Markup, LaTeX)
  - We will use university-internal AI-tools to help us provide feedback

Exam and Course Work

- Exam date: **03.03.2027; 9h-11h** (see HisInOne)
  - You need to register for the exam (“Prüfung”) and evtl. also course work (“Studienleistung”, if required by your study curriculum) in the official time window[1]
  - Course work (“Studienleistung”) registration applies **only for Master students**
    - The reverse is also important: If you should decide to de-register from the exam, also de-register from the course-work.
  - The course work is passed by passing the written exam. Exercise submissions are voluntary
- Format is a **closed book** written exam
- Old exams found in ILIAS
- A mix of know-how, calculative, and theoretical questions

[1] https://www.tf.uni-freiburg.de/de/lehre/a-bis-z-studium/pruefungen